SCHOOL OF
COMPUTER SCIENCE

BUILD4भारत
HACKATHON 9.0

HACK FOR HUMANITY, BUILD FOR COMMUNITY.

Technical Partner

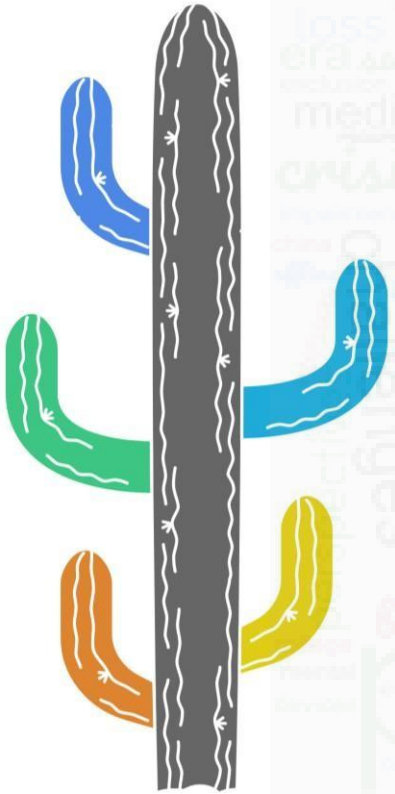
TEAM NAME- THE ENDURANCE

TEAM LEADER NAME- AKSHAT GUPTA

PROBLEM STATEMENT- ASSISTIVE TECHNOLOGY FOR INDEPENDENT LIVING

Sparsh Gyân

Education Barriers for Visually Impaired Students



Social Isolation

Heightens feelings of loneliness, anxiety

Lack of Materials

Limits access to learning resources

Inaccessible Technology

Limits access to online content

Increased Parental Burden

Parents lack training and resources

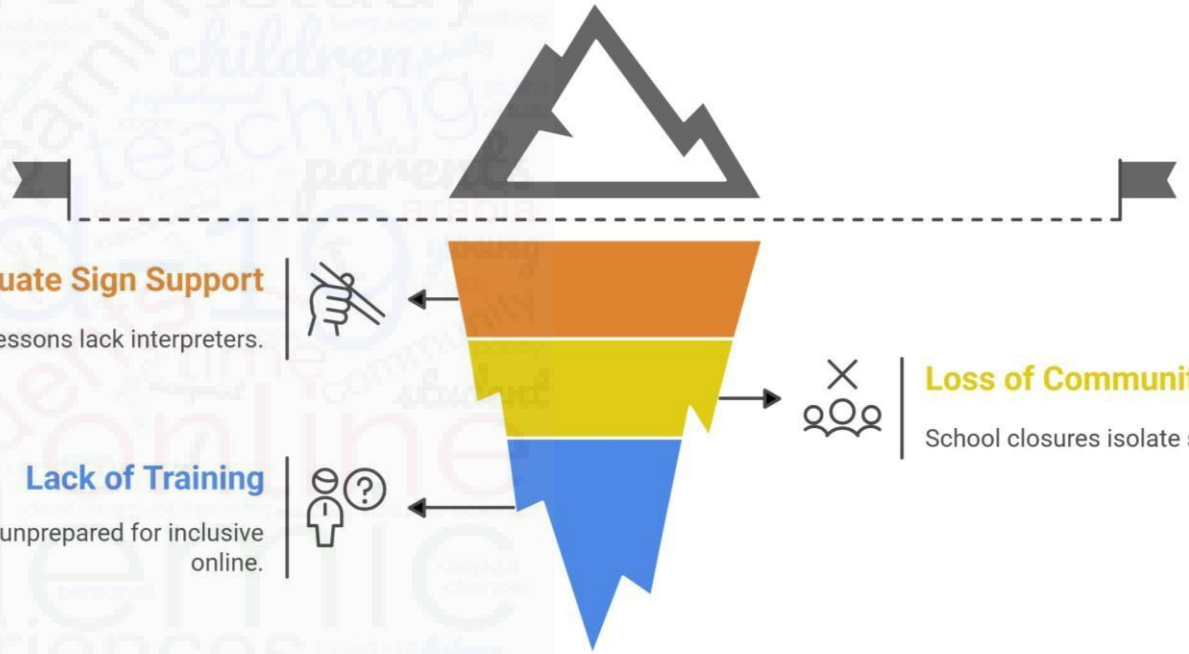
Loss of Hands-On

Hinders tactile and practical learning

Challenges for DHH Students During Pandemic.

Communication Barriers

Face masks hinder lip-reading.



Inadequate Sign Support

Online lessons lack interpreters.

Lack of Training

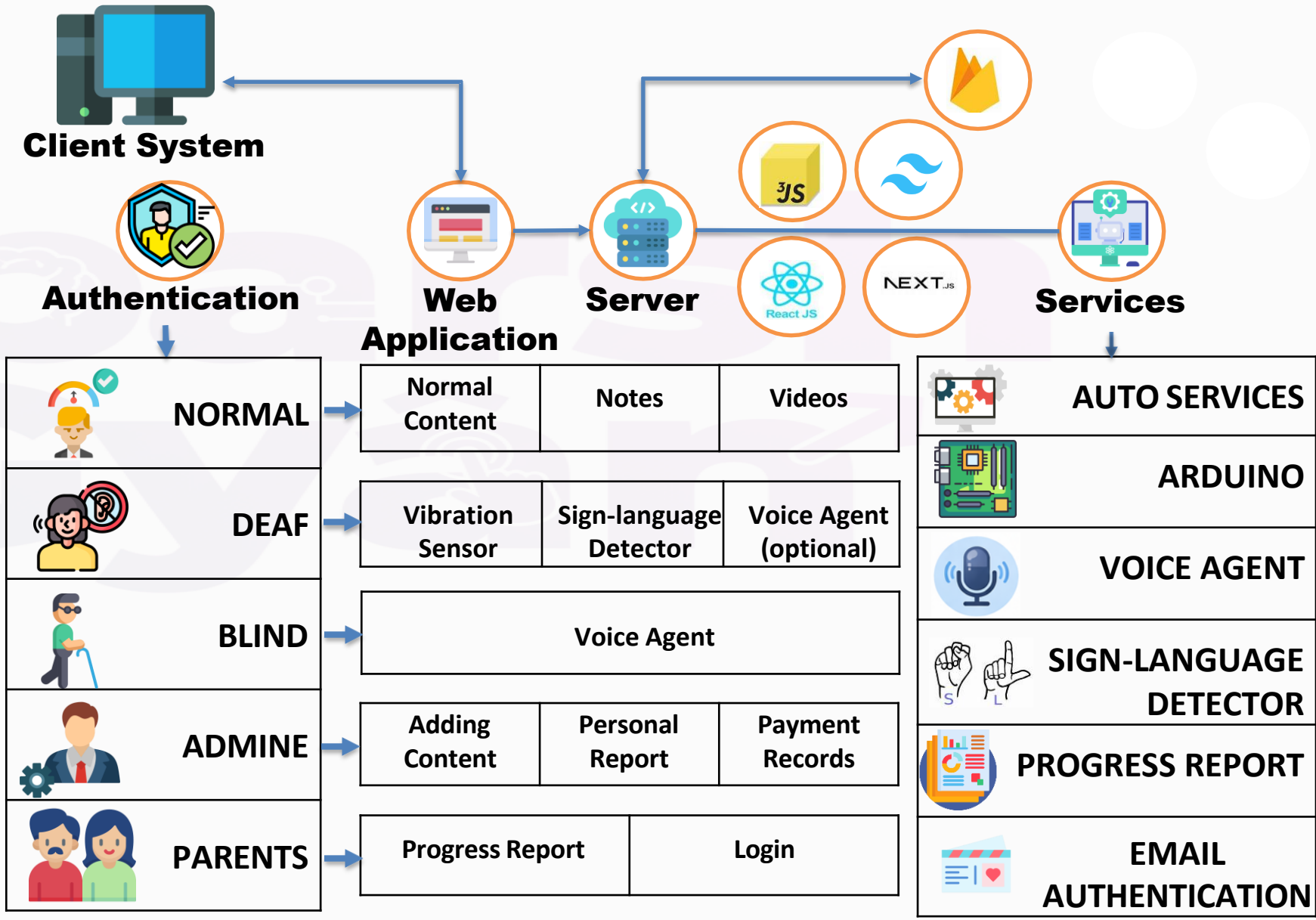
Teachers unprepared for inclusive online.

Loss of Community

School closures isolate students.

PROPOSED SOLUTION

- **Unified Accessibility Hub:** A comprehensive web platform catering simultaneously to visual, hearing, and physical impairments.
- **Hardware-Bridged Braille Output:** Real-time translation of digital on-screen text into physical Braille displays via Arduino integration.
- **AI-Powered Gesture Navigation:** Touch-free computer interaction using advanced computer vision for a virtual mouse and gesture control.
- **Intelligent Voice Assistance:** Embedded conversational AI and high-quality speech synthesis for seamless, hands-free auditory navigation.
- **Real-Time Visual Context:** Live environmental captioning and visual analysis to assist visually impaired users in understanding their surroundings.
- **Audio-First Interactive UX:** Instant auditory onboarding and feedback cues designed specifically to guide differently-abled users from the moment the site loads.



Technical Approach

🚀 **Technology Stack:**

- **Frontend:** React.js + Tailwind (Accessible UI)+ Next.js+ Three.js.
- **Backend:** Firebase.
- **Accessibility:** TTS, Speech-to-Text, Sign-Language AI Avatar, Arduino, Tactile Motion Vibrator.
- **AI:** Elevenlabs, Native Browser Based API, LLM: Gemini 2.5, Tool calling.
- **Low Level OS:** Forked Open Source Repos.

Impact and Benefits

Empowering Visually Impaired Learners

Dual-Sensory Learning

Enhances learning through combined audio and visual experiences.

Universal Access

Integrates with MOOCs to provide broad educational resources.

Independence

Enables autonomous navigation through voice and gesture control.

Expanding Tactile Learning

Basic Text Literacy

.imited to basic text and simple shapes

Subject Expansion

Advanced STEM and tactile diagrams

Enhanced Learning

Tactile games and immersive integration

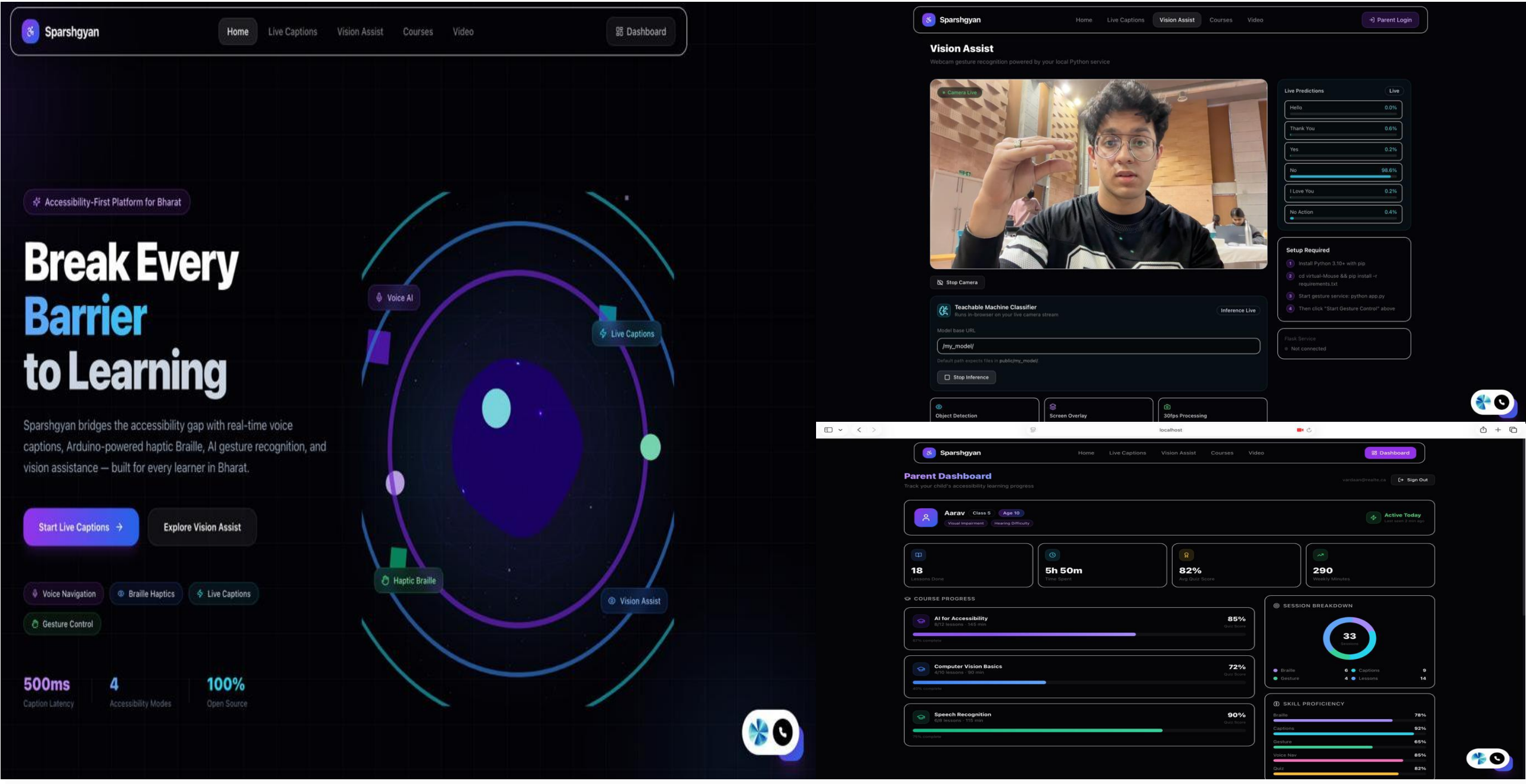
Strategic Deployment

Partnerships for pilot programs

Global Tactile Education

Accessible STEM for all learners

Prototype


Sparshgyan
Home
Live Captions
Vision Assist
Courses
Video
Dashboard
Accessibility-First Platform for Bharat
Break Every Barrier to Learning
Sparshgyan bridges the accessibility gap with real-time voice captions, Arduino-powered haptic Braille, AI gesture recognition, and vision assistance — built for every learner in Bharat.
Start Live Captions
Explore Vision Assist
Voice Navigation
Braille Haptics
Live Captions
Gesture Control
500ms
Caption Latency
4
Accessibility Modes
100%
Open Source
Voice AI
Live Captions
Haptic Braille
Vision Assist
Sparshgyan
Home
Live Captions
Vision Assist
Courses
Video
Parent Login
Vision Assist
Webcam gesture recognition powered by your local Python service
Camera Live
Stop Camera
Live Predictions
Live
Hello 0.0%
Thank You 0.6%
Yes 0.2%
No 98.6%
I Love You 0.2%
No Action 0.4%
Setup Required
1 Install Python 3.10+ with pip
2 cd virtual-Mouse && pip install -r requirements.txt
3 Start gesture service: python app.py
4 Then click "Start Gesture Control" above
Flask Service
Not connected
Teachable Machine Classifier
Runs in-browser on your live camera stream
Inference Live
Model base URL
/my_model/
Default path expects files in public/my_model/
Stop Inference
Object Detection
Screen Overlay
30fps Processing
localhost
Sparshgyan
Home
Live Captions
Vision Assist
Courses
Video
Dashboard
Parent Dashboard
Track your child's accessibility learning progress
Sign Out
Aarav
Class 5
Age 10
Visual Impairment
Hearing Difficulty
Active Today
18
Lessons Done
5h 50m
Time Spent
82%
Avg Quiz Score
290
Weekly Minutes
COURSE PROGRESS
AI for Accessibility
85%
Computer Vision Basics
72%
Speech Recognition
90%
SESSION BREAKDOWN
33
Braille 6
Captions 9
Gesture 4
Lessons 14
SKILL PROFICIENCY
Braille 78%
Captions 92%
Gesture 65%
Voice Nav 85%
Quiz 82%

RESEARCH AND REFERENCES

❖ REFERENCES

- uDot → https://www.udot.tech/
- VHS Learning Accessibility page → https://www.vhslearning.org/about-us/accessibility
- Irish Deaf Society on Brightspace → D2L's page for IDS → https://www.d2l.com/en-mea/why-d2l/customers/irish-deaf-society/
- VO Academy → https://www.voacademy.net/
- Deaf Learn Digital LMS (Deaf-in-Tech) → https://deafintech.org/deaf-learn-digital-lms/

Comparison With Existing Systems

FEATURE	Sparshgyan	uDot	VO Academy	Deaf-in-Tech
Real time monitoring	✔	✘	✘	✔
Transparency impact	✔	✔	✘	✘
Gamifies Engagement	✔	✘	✔	✘
Ai Driven System	✔	✘	✘	✔
Personalized Mentoring	✔	✔	✘	✘
Community Building	✔	✘	✘	✔
Alumni Directory	✔	✔	✘	✘
Centralized Management System	✔	✘	✔	✔